

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE
Supplementary End Sem Examination- Summer 2024

Course : B.Tech

Semester: III

Branch: Artificial Intelligence and Data Science Engineering and Allied branches

Subject Name : Digital Logic and Signal Processing

Subject Code: BTAIES305_Y23 / BTESC305_Y22

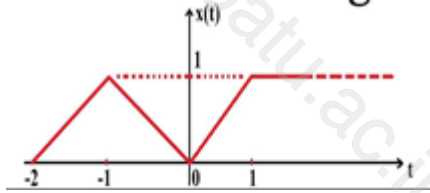
Total Marks: 60

Date: 09/07/2024

Duration: 3 Hrs

Instructions to the students:

1. All questions are compulsory.
2. Assume suitable data wherever necessary.

Q.1 Attempt any two of following.	Level/CO	Marks
1. Draw and explain following logic gates i)AND ii)EXOR iii)OR	CO1/L2	6
2. Perform the following operation i)11001 x 1011 iii)100100 ÷ 111	CO1/L3	6
3. Convert i) (65) _H into octal, binary number system ii) (17) _O into Hexadecimal and binary number system	CO1/L3	6
Q.2 Attempt any two of following.		
1. Sketch following signals for x(t) given as shown below i)x(3t) ii) x(t-2) iii)x(1-t)	CO2/L3	6
		
2. Check the signal is periodic or not and find fundamental period if periodic. i)X(t)=2sin5πt+5cos(2πt) ii) x(n)=2cos5n	CO2/L3	6
3. Check the signal is time variant or time invariant i)y(t) = x(-t) ii) y(t) = t. x(t)	CO2/L3	6
Q.3 Attempt any two of following.		
1. Perform linear convolution using graphical method: x(n)={1,0,-1,3} h(n)={1,2,3,2} where x(0)= -1, h(0)=1	CO3/L3	6
2. Perform circular convolution of x(n)={1,2,1,2} h(n)={ 1, 1, 1, 1}	CO3/L3	6
3. Find 4 point DFT of x(n)={1,2,3,4}	CO3/L3	6
Q.4 Attempt any two of following.		
1. Find inverse z transform of $X(Z) = \frac{1}{1-0.8z^{-1}+0.12z^{-2}}$ using power series method.	CO4/L3	6
2. Find z transform and state ROC of following sequence: i)x(n)=0.5 ⁿ u(n-1) ii) x(n)= u(n)	CO4/L3	6
3. State and prove any two properties of z transform.	CO4/L3	6
Q.5 Attempt any two of following.		
1. State properties of autocorrelation.	CO5/L2	6
2. State properties of energy spectral density.	CO5/L2	6
3. Find auto correlation of x(n)={1,2,2,1}	CO5/L3	6

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